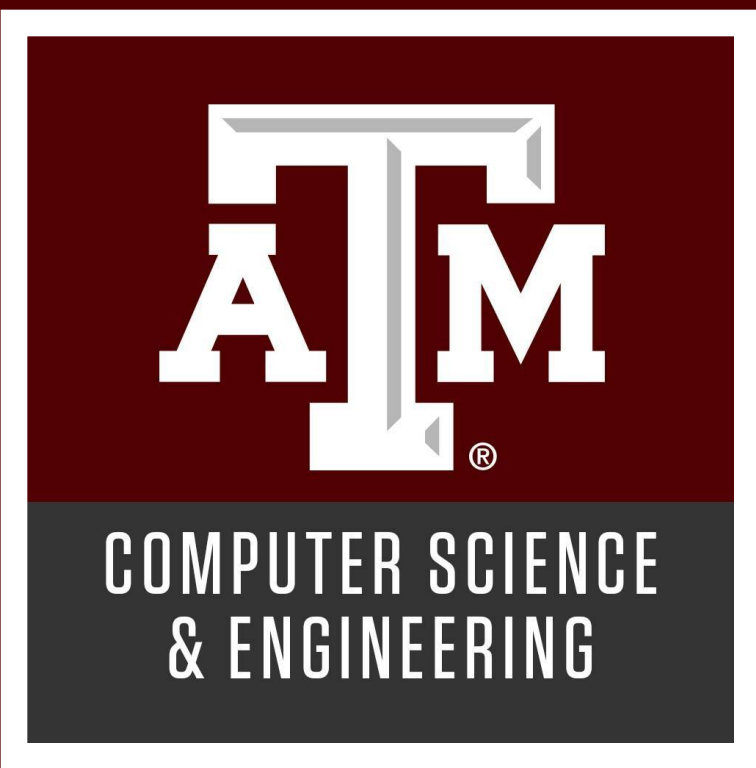




Predictive Modeling in Cow-Calf Operations Using a Univariate Modeling Framework for Economic and Sustainability Insights

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INTRODUCTION

- U.S. cow-calf operations make up nearly half (47.7%) of the national cattle inventory, with feed costs often exceeding 50% of total expenses..
- High variability from weather, fertility, and market fluctuations makes economic performance difficult to predict using traditional methods.
- Time-series models can enhance forecasting accuracy, offering a way to capture seasonal and cyclical patterns in cow-calf market indicators.

OBJECTIVES

- Evaluate univariate time-series models (ARIMA, SARIMA) for forecasting calf prices, capturing seasonality and volatility.
- Extend forecasts with SARIMAX models incorporating economic indicators such as feed costs, fuel prices, CPI, and exchange rates.
- Demonstrate how machine learning-based forecasting can support decision-making and promote sustainability in cow-calf operations.

DATA & FEATURE ENGINEERING

- Monthly data from the USDA, spanning January 1996 through December 2024
- Contains Date (first of each month), Gross Revenue (target), Net Gas Price, Corn Price, CPI, USD-JPY Exchange Rate
- Gross Revenue was adjusted for inflation using CPI
- Real Gross Revenue was differenced once to remove seasonality and achieve stationarity
- An 80/20 train-test split was used in forecasting

Full dataset

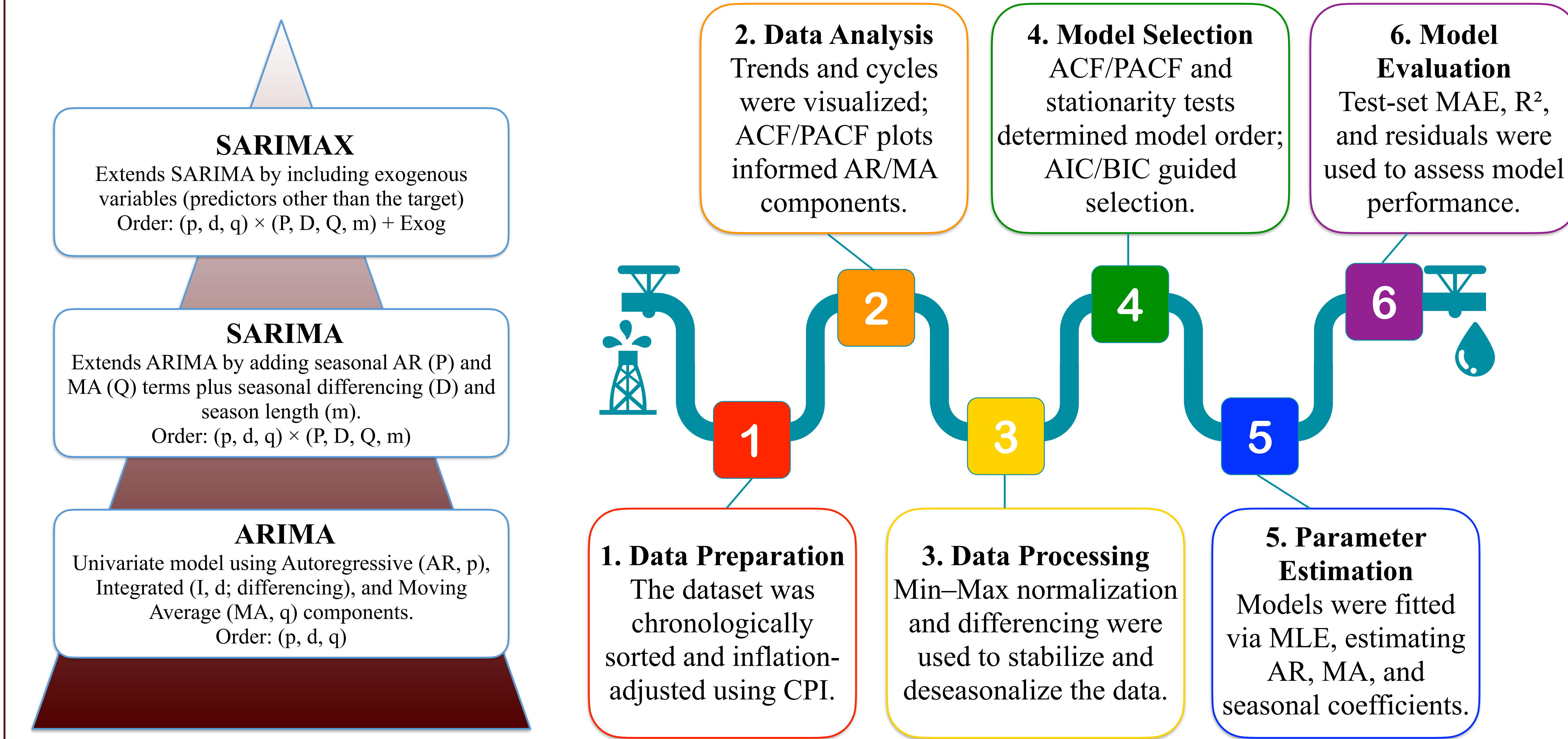


REFERENCES

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References



METHODOLOGY



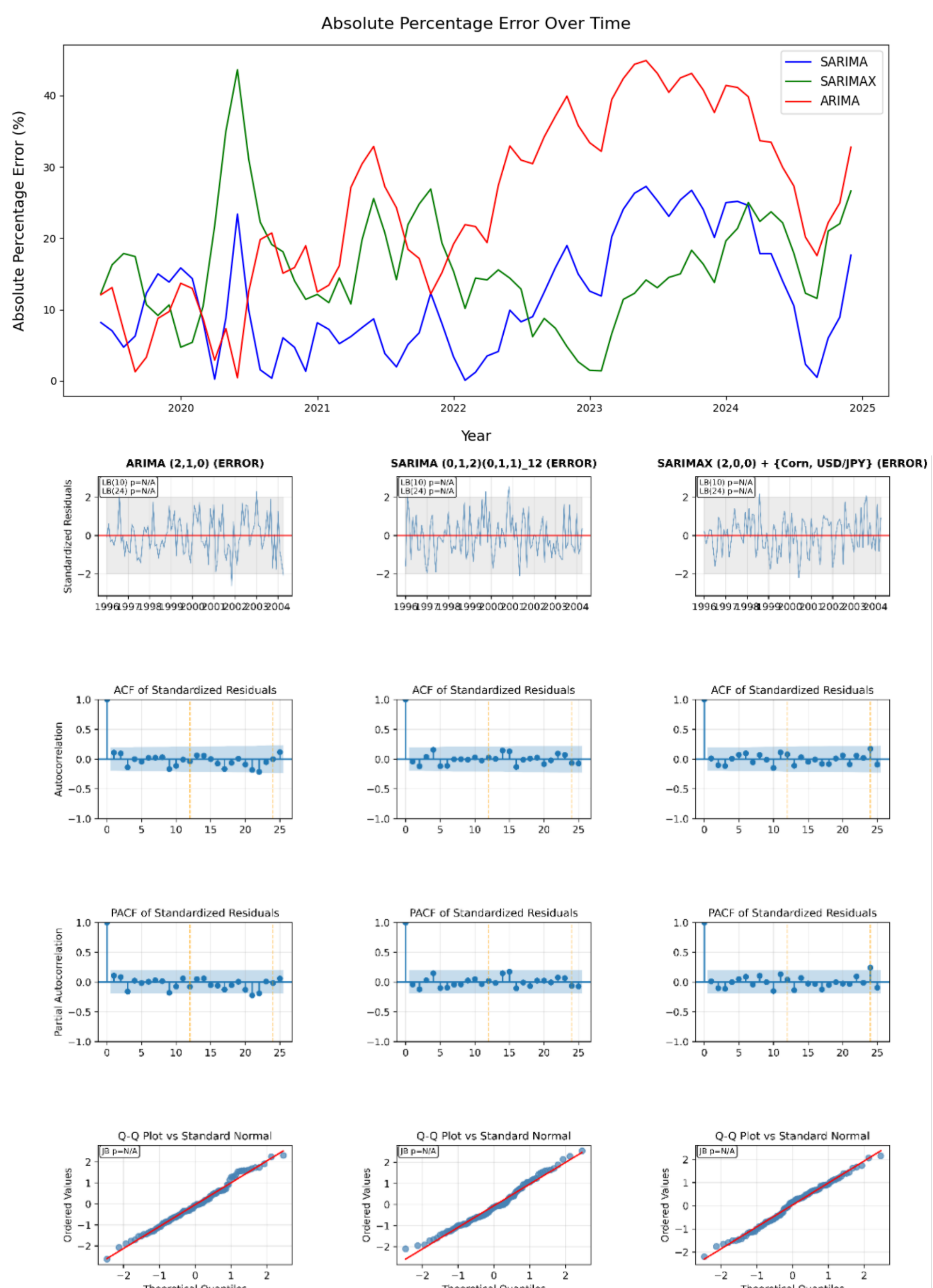
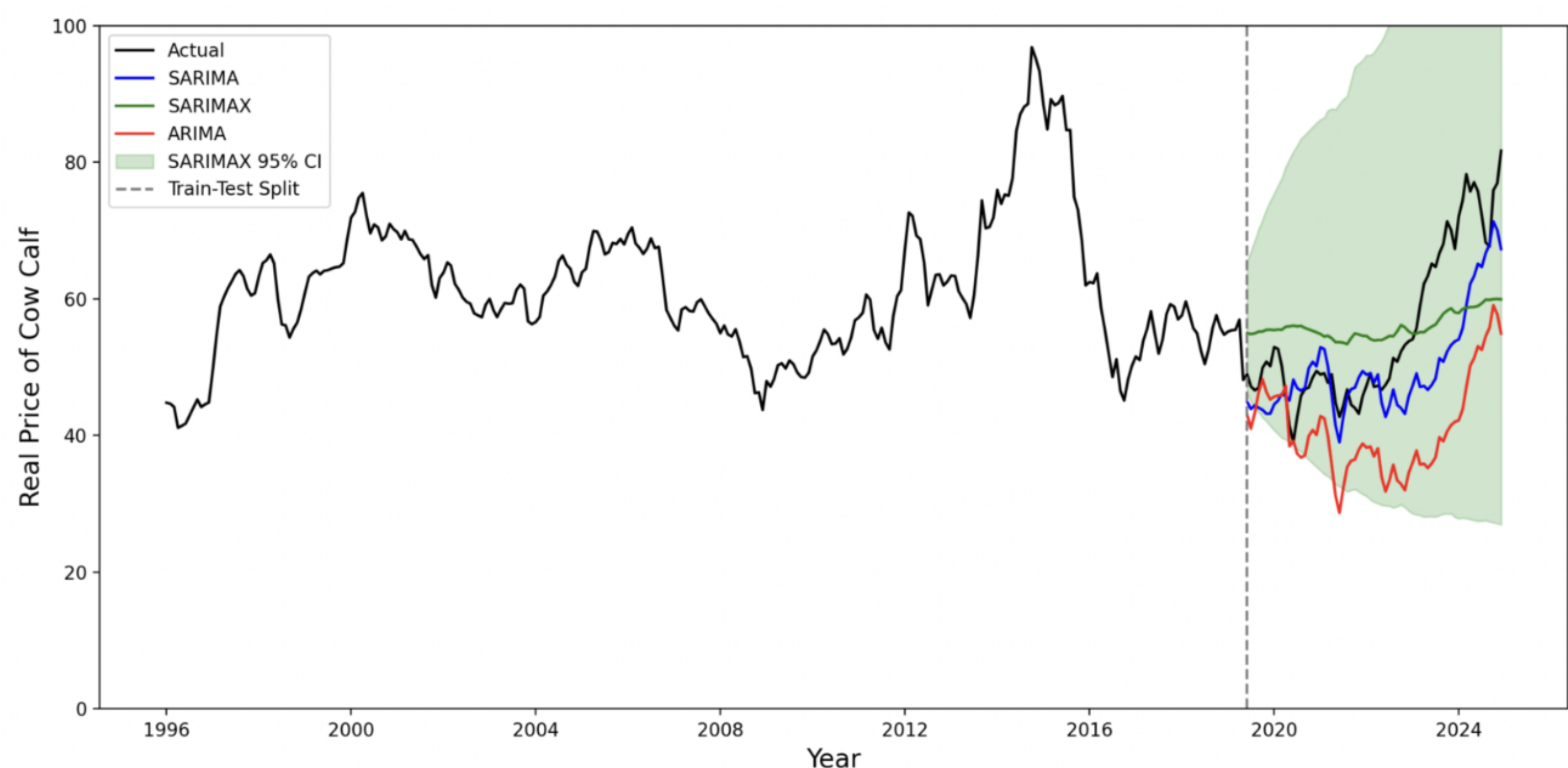
Model Assumptions:

- Data is stationary (constant mean & variance)
- Relationships are linear
- Residuals are uncorrelated and homoscedastic (constant variance)
- Model errors exhibit no autocorrelation after fitting
- Seasonal patterns are stable and repeat consistently across periods (SARIMA, SARIMAX)
- Exogenous variables are relevant, not collinear (high correlation between two exogenous variables), and independent model residuals (SARIMAX)
- Parameters remain constant over time

Model Limitations:

- Accuracy decreases over longer forecast horizons
- Structural breaks or sudden shifts due to policy changes

RESULTS & ANALYSIS



Model	MAE	RMSE	MAPE	MPE	RMSPE	R ²
ARIMA (2,1,0)	14.50	17.10	24.63%	24.53%	27.55%	-1.21
SARIMA (2,1,0) (0,1,1) ₁₂	6.99	9.04	11.67%	8.16%	14.15%	0.38
SARIMAX (2,0,0) (0,0,0) ₁₂ + Corn, USD/JPY	8.66	10.83	14.97%	14.80%	17.94%	0.13

CONCLUSIONS

- Explicitly modeling seasonality with SARIMA produced the highest accuracy, outperforming both SARIMAX and ARIMA.
- SARIMAX's slight performance decrease compared to SARIMA suggests that our chosen external regressors may not be able to fully explain additional variance.
- Seasonal patterns remain the dominant driver of calf price variation, reflecting consistent annual trends driven by production cycles, breeding patterns, and market demand.

NEXT STEPS

- Develop and evaluate multivariate machine learning models such as Random Forest, LightGBM, and other supervised learning models to capture complex interactions among predictors and improve forecasting accuracy.
- Expand exogenous features to include market indicators, weather patterns, and relevant policy.
- Combine time-series and machine learning models to further improve performance.
- Explore model refinements to improve univariate forecasting beyond current approaches.